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Integrated experimental and simulation approach to establish the effect of elemental segregation in Inconel 718 welds

Rahul M R [a,*], M. Agilan [b], Dasari Mohan [c,d], Gandham Phanikumar [c]

[a] Department of Fuel Minerals and Metallurgical Engineering, Indian Institute of Technology (ISM) Dhanbad, Jharkhand 826004, India
[b] Materials and Metallurgy Group, Vikram Sarabhai Space Centre, ISRO, Thiruvanthapuram 695022, India
[c] Department of Metallurgical and Materials Engineering, Indian Institute of Technology Madras, Chennai, Tamil Nadu 600036, India
[d] Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Bombay, Mumbai, Maharashtra 400076, India

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ABSTRACT

Inconel 718 as-welded structures are widely used in many critical applications. Welding will cause different types of detrimental phase formation and segregation of elements in the inter-dendritic region. The segregation of elements depends on the local curvatures during morphological evolution dictated by the thermal conditions prevailing during the process. In this study, we establish the segregation behavior of Inconel 718 as a function of solidification conditions. Microstructure evolution was correlated with Scheil's solidification predictions for different cooling rates and temperature gradients. The solidification pathway of segregated composition depends on the process parameters via the microstructure. Simulation results are validated with the experimental results, and the microstructure comparison shows good agreement. An integrated workflow is proposed to accelerate the welding process optimization and produce desired microstructure for multi-component alloys.

1. Introduction

Inconel 718 is an age-hardenable superalloy extensively used in high-temperature applications. Welded structures of Inconel 718 are largely used in many strategic sectors, including aerospace, nuclear, and petrochemical industries. Sonar et al. [1] shown the applicability of inter pulsed TIG welded Inconel 718 components for the advanced high power jet engines. Kwon et al. [2] shown the cryogenic properties of electron beam welded Inconel 718 for aerospace components. Several reports available in literature about the characteristics of Inconel 718 welds used in aerospace industries [3–7]. The effective use of the alloy in any application will depend upon the microstructure stability and mechanical properties. Inconel 718, containing alloy elements such as Nb, Ti, Mo, etc., can cause segregation depending upon the processing conditions. This segregation may cause the formation of detrimental phases and affect precipitation behavior. Mantri et al. [8] reported the formation of γ' and γ" in the additive manufacturing sample and showed that chemical inhomogeneties (micro-segregation) would affect precipitation behavior. Thavamani et al. [9] used an ultrasonic vibration technique in TIG welding of Inconel 718 to reduce hot cracking behavior, confirming the effect of solute segregation in hot cracking. The recent literature suggests that depending on the process such as laser [10], TIG welding, etc., the inter-dendritic region is characterized by the formation of precipitates or phases [7]. Agilan et al. studied the effect of heat input on elemental segregation in the dendritic and inter-dendritic regions of Inconel 718 electron beam welds [11]. Anbarasan et al. [12] reported the effect of shielding gas and filler material for Laves phase formation by restricting Nb segregation. Zhao et al. [13] reported that the time duration for homogenization of electron beam processed IN718 will depend on the segregation of elements and morphological characteristics. Wang et al. [14] show that ultrasonic arc welding will modify the Laves phase amount and morphology. The mechanical property correlation with the process is reported in the literature [15–17]. The segregation behavior in Inconel 718 can thus be said to affect the phase formation, precipitation behavior, and mechanical properties.

The micro and macro-scale modeling approaches can identify the process and microstructure correlations. Literature suggests that microstructure modeling will help avoid detrimental phase formation [18]. The recent report of the FEM simulation of the TIG welding process in Inconel 718 shows the cooling rate variation from 0 to 900° C/s [19]. Lin et al. [6] used a FEM model to establish the stress evolution in laser-welded Inconel 718 alloy during cooling. Kumara et al. [20] reported the application of phase-field modeling to establish the segregation pattern of superalloy in additive manufacturing conditions. The segregation behavior is dependent on the process parameters via the microstructure. Understanding segregation behavior at different processing conditions will help identify the optimum processing window. From the previous reports, it is clear that an integrated framework is necessary to identify

* Corresponding author.
E-mail address: rahulmr@iitism.ac.in (R.M. R).

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Table 1
Parameters used for phase-field simulation in the current study.

Parameters required for phase-field simulation	Value
Kinetic coefficient (μ) between phases	4.00×10^{-02} [$cm^4/(Js)$]
Surface energy between phases	1.00×10^{-05} [J/cm^2]
Domain size, Grid size	200 × 200 μm, 0.25 μm with moving frame in z direction

the optimum condition to avoid or reduce the detrimental phase formation of Inconel 718 while welding and solidification. In the current study, the segregation pattern during TIG welding was established using phase-field simulation, the phase formation in the segregated composition was analyzed using thermodynamic and kinetic simulation. The experimental and simulated compositional data were compared. Continuous cooling transformation (CCT) diagram was generated based on the segregated composition. An iterative integrated framework is proposed for welding process optimization.

2. Experimental details

Gas tungsten arc welding was performed on Inconel 718 sheets of 5 mm thickness using Kemppi make (Model No: Mastertig AC/DC 3500W) welding machine. The welding parameters were 120 A current, 12 – 13 V voltage, and 90 mm/min welding speed. K-Type thermocouples were placed in the fusion zone and heat-affected zone to acquire the temperature profile in these regions. Temperature data was acquired using Yokokawa (Model MW100) make data acquisition unit. The welded sample is sectioned and polished using standard metallographic polishing methods. Elemental segregation between the dendritic and interdendritic regions of welds was analyzed using an energy dispersive X-ray spectrometer (EDS) attached with the scanning electron microscope (SEM), Tescan Vega 3®.

3. Simulation details

The current study uses thermodynamic simulation using Thermo-Calc® [21] and phase-field simulation using Micress® [22] as well as in-house developed code. Considering the important role of microstructure evolution and subsequent micro-segregation, we compare the results of two different phase-field models here. For calculating Scheil's plot and the amount of fraction of phases vs. temperature plot, the Ni-based thermodynamic database TCNI11 is used. The phase-field simulation in Micress® was performed using a multi-phase-field approach [23] with a TQ interface with Thermo-Calc®. Below is the modified multiphase field equation for the current study

$$\frac{\partial \Phi}{\partial t} = \mu^*(\vec{n})\left[\sigma^*(\vec{n})\left(\nabla^2\Phi + \frac{\pi^2}{\eta^2}\left(\Phi - \frac{1}{2}\right)\right) + \frac{\pi}{\eta}\Delta G\sqrt{\Phi(1-\Phi)}\right] \quad (1)$$

where the phase-field order parameter is represented by Φ, which takes a value for 1 for FCC phase and 0 for liquid, and the solid-liquid boundary will take a value of -1. The interfacial mobility (μ^*) and the anisotropic interfacial energy ($\sigma^*(n)$) parameters are optimized for the current problem. The diffuse interface thickness is represented as η. The mobility database MOBNI5 was used for diffusion data, and diffusion equations incorporated can be found in the literature [24]. The FCC phase and liquid were considered, and constant undercooling was used for all simulations. The optimized kinetic coefficient and surface energy values are used, and these values are kept constant for all simulations for comparing the results. Table 1 shows the values used for the current study. The length scale of the simulation domain allows for a microstructure simulation using constant cooling rate and temperature gradient in the growth direction. The experimental thermal cycle collected from the weld pool is incorporated in the phase-field simulation, and microstructure evolution is captured for comparative purposes. The JMatPro® software [22] with NiFe-based superalloy database was used for the kinetic calculation to generate the CCT diagram to compare the effect of segregation.

An in-house code phase-field model was used to ensure the independence of microstructure simulations on the phase-field model [25]. The phase-field variable denotes the phase: $\phi=0$ in the liquid phase and $\phi=1$ in the solid phase. The solid-liquid interface is represented by values between 0 and 1. The evolution equation for the phase-field variable is

$$\frac{1}{M_\phi}\frac{\partial \phi}{\partial t} = \varepsilon_\phi^2\nabla^2\phi - W\frac{\partial g(\phi)}{\partial \phi} - \frac{\partial h(\phi)}{\partial \phi}\left(f^S - f^L - \sum_{i=1}^{n}\left(c_i^S - c_i^L\right)\tilde{\mu}_i\right) \quad (2)$$

and evolution equation for composition is

$$\frac{\partial c_i}{\partial t} = \nabla\cdot(1-h(\phi))\sum_{j=1}^{n} D_{ij}^L\nabla c_j^L + \nabla\cdot j_{at_i} \quad (3)$$

where M_ϕ is phase-field mobility, ε_ϕ is gradient energy coefficient, $g(\phi)$ is the double-well function, W is the height of double-well function, $h(\phi)$ is interpolation function, f^S and f^L are free energy functions for solid and liquid respectively, c_i^S and c_i^L are compositions in solid and liquid, respectively, $\tilde{\mu}_i$ is chemical potential and D_{ij}^L the diffusion coefficient in liquid. The subscript i represents the ith component. The term j_{at_i} represents the antitrapping current introduced in the model and given as

$$j_{at_i} = \frac{\varepsilon_\phi}{\sqrt{2W}}\left(c_i^L - c_i^S\right)\frac{\partial \phi}{\partial t}\frac{\nabla\phi}{|\nabla\phi|} \quad (4)$$

A detailed description of the model parameters can be elsewhere in the literature [26]. The temperature gradient was imposed in the direction of the growth (Y-axis) is given by

$$T(y) = T_{int} + T_G(y - Vt) \quad (5)$$

where, T_{int} is the temperature at $t = 0$ and position y, T_G is the temperature gradient along the growth direction, and V is growth velocity.

The code is developed in C-programming language, and a parallel programming approach based on MPI [27] libraries and GPU programming is adopted. MPI+OpenCL [28] approach is implemented on multiple GPUs. Euler explicit scheme is employed for solving governing equations. Periodic boundary conditions are employed for domain boundaries parallel to growth direction, and Neumann (no flux) boundary conditions are implemented for the boundaries perpendicular to the growth direction. CALPHAD [29,30] based descriptions for Gibbs energies are incorporated into the phase-field code to describe the actual alloy systems. We considered a ternary Ni-Cr-Nb alloy system for in-house code simulation, and the Gibbs energy parameters are taken from Du et al. [31]. The diffusion coefficients are taken from the MOBNI4 database of Thermo-Calc® software. A domain size of 5120 × 5120 grid points with a grid size of 20 nm is used in the current simulations.

4. Results and discussion

The Inconel 718 alloy is used in many applications where the material is subjected to a thermo-mechanical field and requires welding as a fabrication stage. Based on the welding parameters, the thermal field established in the material will be different, and the melt pool characteristics will be varied [32–34]. The weld microstructure depends on the local thermal conditions. These can be changed via the welding process parameters and boundary conditions. Among such thermal conditions,

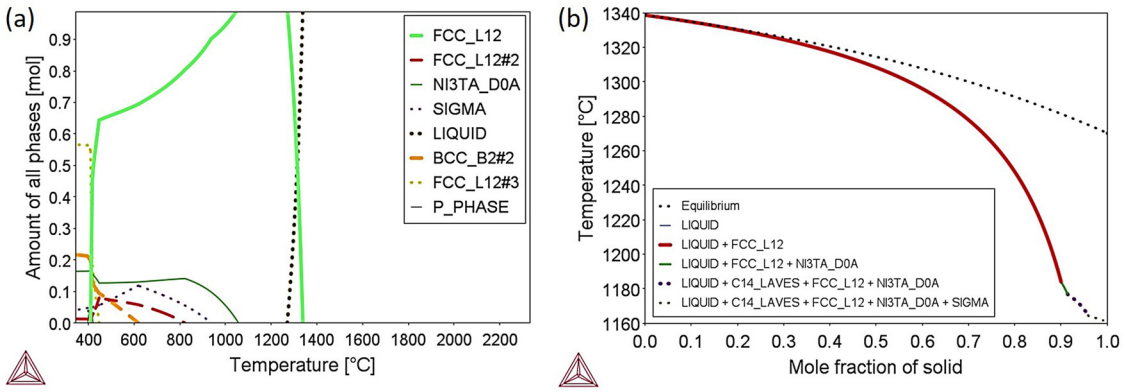


Fig. 1. Thermo-Calc® prediction of phase formation in the current alloy (a) Amount of phase vs. temperature for Inconel 718 alloys, (b) Scheil's solidification plot shows the solidification pathway.

Table 2
Average composition of the base material (composition used for micress simulation).

Element	Al	Cr	Fe	Mo	Nb	Ti	Ni
Composition (wt %)	0.5	20	15.5	3	5	1	Balance

the cooling rate and the thermal gradient are the most important factors that will control the microstructure formation in the weld pool. Fig. 1 shows phase fraction vs. temperature and Scheil's solidification data [35,36] of Inconel 718 alloy selected for the current study, which confirms the phase formation in the alloy. The average alloy composition from EDS for the current study is shown in Table 2. FCC primary phase will form from the liquid, and Scheil's plot confirms that the FCC fraction will be more than 80% which is on par with the literature [9]. Scheil's plot shows a solidification range of ~180 °C and the formation of Laves, sigma, etc., which confirms the probability of hot cracking in the alloy. One could note that beyond a solid fraction of 10%, the inter-dendritic region can have different solidification pathways depending on the local segregated composition.

The phase-field simulations with different thermal gradients and cooling rate combinations were carried out in Micress. The range of thermal gradient and cooling rate for this study was identified from the literature [37,38]. Manikandan et al. [37] confirm that the cooling rate can be altered by varying the welding condition and shielding gas, and the temperature gradient varies in the range of 0 to 1000 K/cm. In the current study, a 4 × 4 matrix was considered for phase field simulation with cooling rate values of 10, 100, 500, 1000 K/s and thermal gradient values of 10, 100, 500, 1000 K/cm. Fig. 2 shows the Nb concentration profile at the growing dendrite as different combinations of temperature gradient and cooling rate. The map shows the FCC+ Laves phase fraction based on the segregated interdendritic composition. The average interdendritic composition was manually extracted from the solute fields resulting from phase-field simulations in Micress. The timestep used for solute field output (data after 0.1 s of solidification) and the location are ensured to be representative for the respective combination of thermal gradient and cooling rate. For each of those segregated compositions, Scheil's solidification plot was generated. From the Scheil's plot the fraction of FCC+Laves and solidification range was identified and plotted as a map. One could note that the continuous contour is obtained by interpolating the values between the points.

In the phase-field simulations, the temperature gradient is given along the growth direction and can be correlated with the dendrite growing from the pool boundary to the weld center. One could note that Nb segregates into the inter-dendritic region, and the extent of segregation depends on the combination of cooling rate and thermal gradient. The morphology of growing dendrite varies with the gradient and cooling rate, which is the product of the growth rate of the dendrite (R) and gradient (G). When the cooling rate increases, the inter-dendritic spacing decreases. While increasing the gradient at the constant cooling rate, the microstructure changes from dendritic to columnar dendritic morphology. This morphology variation is evident at high gradient conditions. Such morphological changes lead to micro-segregations via the curvature effects. The morphological variations and the micro-segregation were compared with the output from the in-house phase-field code.

Fig. 3 (a & b) compare the microstructure simulated using Micress® and developed phase-field code with the same gradient and cooling rate conditions. Fig. 3(b) represents microstructure simulated from in-house phase-field code. Initial rectangular seed evolved with time under temperature gradient 1000.0 K/cm and growth velocity 6.0 mm/s. A dendritic microstructure is developed during growth with secondary and tertiary arms. Cr with partition coefficient greater than one is enriched in the dendrite region, and Nb with partition coefficient less than one is enriched in the interdendritic region. One could note that the morphology and segregation profile and length scale are comparable (Fig. 3 (a&b)), which implies the robustness of the applicability of phase-field to these applications. Fig. 3 (c&d) represents the comparison of segregation profile along a line across the interdendritic region. This shows the segregation profile is comparable in both simulations. The contour map in Fig. 2 confirms the variation of the fraction of FCC+ laves phase as a function of processing conditions. Laves +FCC phase formation in the segregated region will be more than 35% for the cooling rate between 400 to 600 K/s at a shallow temperature gradient of about 200 K/cm. A large domain of less than 15% fraction of FCC+ Laves can be seen in the map, which will be a relatively safe region for welding to reduce detrimental phase formation. Fig. 4 shows the solidification range mapped for the segregated compositions in the interdendritic region as a function of thermal gradient and cooling rate. The solidification range of the interdendritic compositions can vary drastically depending on the process conditions. The minimum solidification range overlaps with the region of the maximum FCC+Laves phase region, which shows the final liquid can result in a eutectic reaction. The diagram also shows a safe region of a low solidification range to avoid hot cracking. One could conclude that by effectively utilizing the maps (Figs. 2 & 4) in the parametric space of thermal gradient and cooling rate, one can control the solidification path in the inter-dendritic region by controlling the welding parameter.

To corroborate the workflow followed in this study, we have performed experimental trials and compared them with the simulation data. Fig. 5 shows the experimental comparison with the phase-field simulation. EDS map shown in Fig. 5(a) was taken from the thermocouple location in the fusion zone shows the segregation of Mo, Nb, Ti, and depletion of Fe, Cr in the inter-dendritic region. The thermal cycle

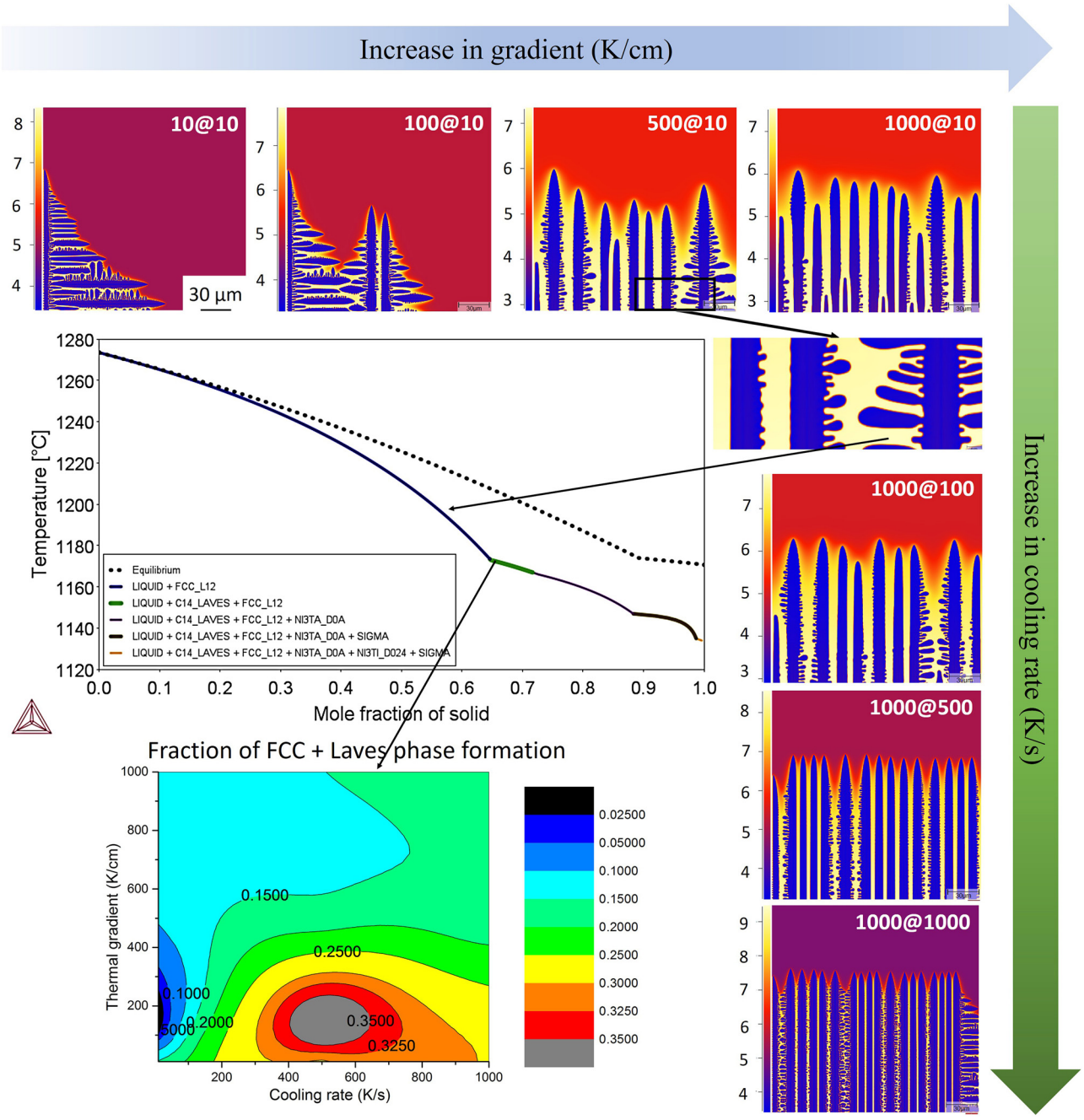


Fig. 2. Tip of growth dendritic front show the segregation of Nb with respect to cooling rate and gradient, contour map shows the Scheil's prediction of the fraction of FCC+Laves phase formation for the segregated composition in the inter-dendritic with respect to the cooling rate and thermal gradient (where 500@10 denotes a thermal gradient of 500 K/cm and cooling rate of 10 K/s).

collected from the weld pool is shown in the overlay map inset, and this thermal cycle is incorporated in the phase-field simulation for microstructure simulation. Fig. 5(b) shows the segregation map collected from the phase-field simulation. It shows the segregation of Mo, Nb, Ti, and Al in the inter-dendritic region and enrichment of Cr, Fe in the dendritic region. The inter-dendritic spacing, also comparable with the experimental results, confirms the applicability of simulation results for welding process optimization. The cooling cycles in the weld pool depend on the welding parameters, which can be controlled. Thus, the phase formation maps can identify processing conditions for the desired segregation behavior. The recent literature shows the welding process modification to alter or reduce the segregation effects [12,14,39], and this study will help further optimize the process.

The EDS line profiles across the inter-dendritic regions are shown in Fig. 6 (a&b) for the experiment and simulation, respectively. These confirm the segregation of Mo, Nb, and Ti in the inter-dendritic region. The virtual EDS line scan is comparable to the experimental one. The virtual EDS line profile shows the composition variation within the FCC dendritic phase. A decrease of Nb, Mo near the inter-dendritic region will cause a change in precipitation nature in the dendritic phase. The nature of precipitation will differ in the dendrite core, dendrite periphery, and the inter-dendritic region as reported for laser processing [8]. Scheil's solidification plot (Fig. 6(c)) using the inter-dendritic composition confirms the formation of Laves and sigma phases. The CCT diagram was plotted based on the composition of dendrite and inter-dendritic region respectively is shown in Fig. 6 (d&e). The CCT diagram confirms the

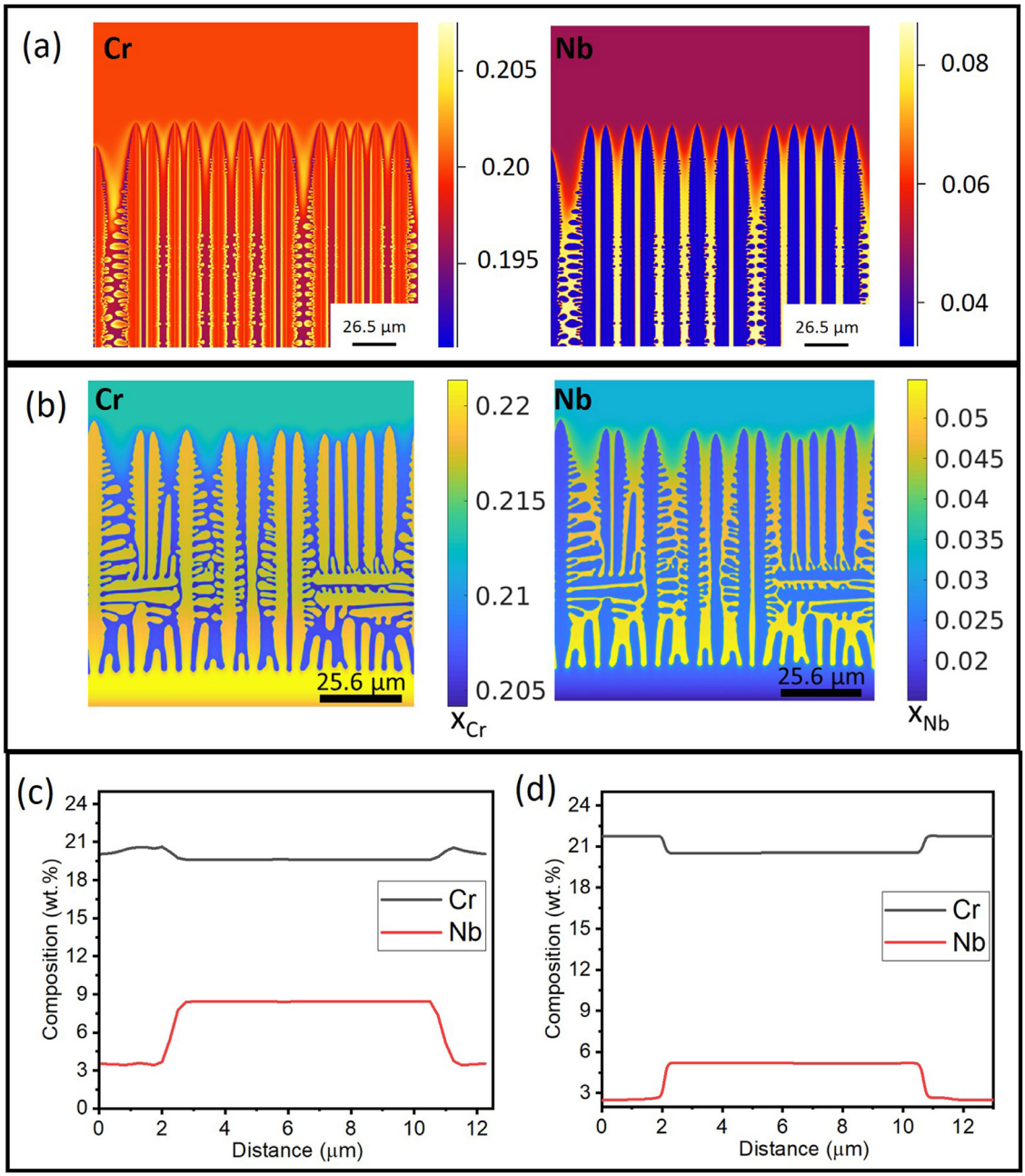


Fig. 3. (a): The map shows the distribution of Cr and Nb with a cooling rate of 600 K/s and thermal gradient 1000 K/cm using the micress simulation, (b) Distribution of Cr and Nb with a cooling rate of 600 K/s and thermal gradient 1000 K/cm using the developed phase-field code (Composition considered - Ni - 21.3 at.% Cr - 3.2.% Nb, colour code shows in atomic fraction), (c) Elemental distribution (EDS Line profile) taken at inter dendritic region of using Micress simulation, (d) Elemental distribution (EDS Line profile) taken at inter dendritic region of using developed phase-field code.

change in precipitation or other phase formation while moving from the dendritic core to the inter-dendritic region. While quantitative predictions may not be accurate, one can infer the probability of phase formation. At the dendrite core, there is a greater probability of forming the phases γ', γ" & delta. Similarly, there is a lesser probability of forming Laves or sigma phases. The CCT diagram at the inter-dendritic region (Fig. 6e) shows the probability of γ', γ", delta, and other phases such as Laves and sigma. One could note that the Laves phase curve is above than γ', γ" curves at a slow cooling rate.

We propose a workflow to extend the integrated approach to other alloys and similar processes. Traditionally, many studies have attempted to directly relate the microstructure with a process condition and the initial alloy composition (as shown using dotted lines in Fig. 7). This approach ignores the microstructure formation and the role of local curvatures. Also, micro-segregations offer a predictive capability for the microstructure or design of alloys. Through *in-situ* diagnostics or calibrated process simulations, one could extract the thermal gradient and cooling rate at locations of importance. These parameters combined with the initial alloy composition can be used as inputs for a simulation of local microstructure using the phase-field approach to provide the micro-segregation. Enriched compositions in the inter-dendritic regions lead to intermetallic phase formation.

The composition of the last liquid to solidify can be seen as a function of thermal gradient and cooling rate. This segregated alloy composition could be used in place of the initial alloy to predict phase formation using Scheil curves. Since the curvature effects due to local microstructure

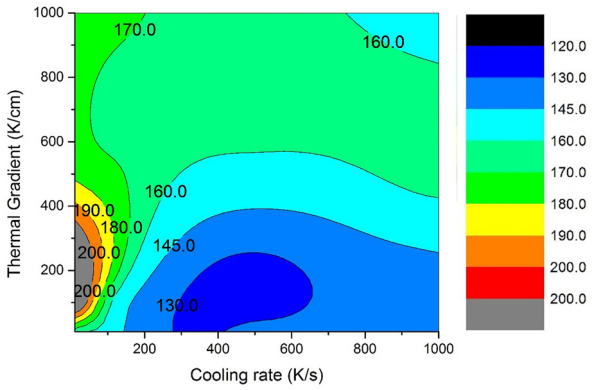


Fig. 4. Contour map shows the solidification range for the segregated composition in the inter-dendritic with respect to the cooling rate and thermal gradient.

have already been considered by phase-field simulation, the segregated alloy is a better predictor than the initial composition. This composition can also be used in time-temperature-transformation (TTT) diagrams to comment on the nucleation and growth of relevant phases. Phase maps can be generated using these calculations and serve as microstructure predictors. Iterations in process conditions can then identify appropriate ranges in the thermal gradient and cooling rate parameters to obtain the desired combination of phases and their fractions. Correlating the thermal gradient and cooling rate to the process is a standard inverse problem in heat transfer. This approach can also be used to rationalize gradients observed in microstructures. We believe that an integrated approach would be useful in designing alloys for emerging processes such as additive manufacturing [40], where thermal gradients and cooling rates could be tuned to be location dependent as required.

5. Conclusion

An integrated microstructure evolution study using simulation and experimental trials was carried out. The segregation of elements in the inter-dendritic region of welded Inconel 718 will alter the local solidification pathway. The elemental distribution within the dendrite and inter-dendritic region will alter the precipitation behavior. The microstructure simulations take local curvature effects and offer micro-segregation profiles that affect the phase formation. The contour maps of phase fraction formation and solidification range were established with gradient and cooling rate. These studies will help to accelerate the welding process optimization with fewer experimental trials.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

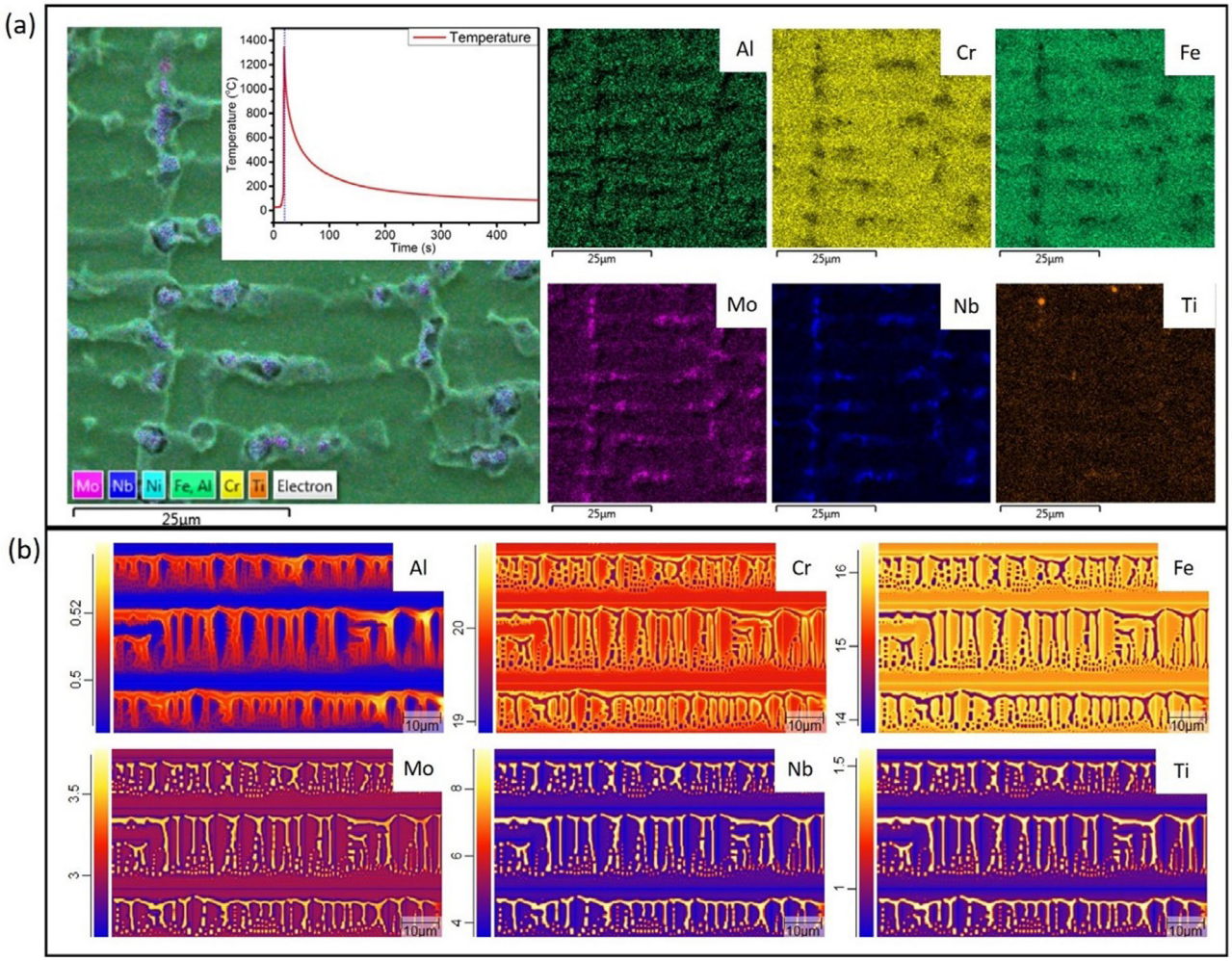


Fig. 5. Experimental validation, (a) Experimental EDS mapping and individual elemental segregation map (inset shows the thermal profile collected from fusion zone), (b) Simulated elemental segregation map.

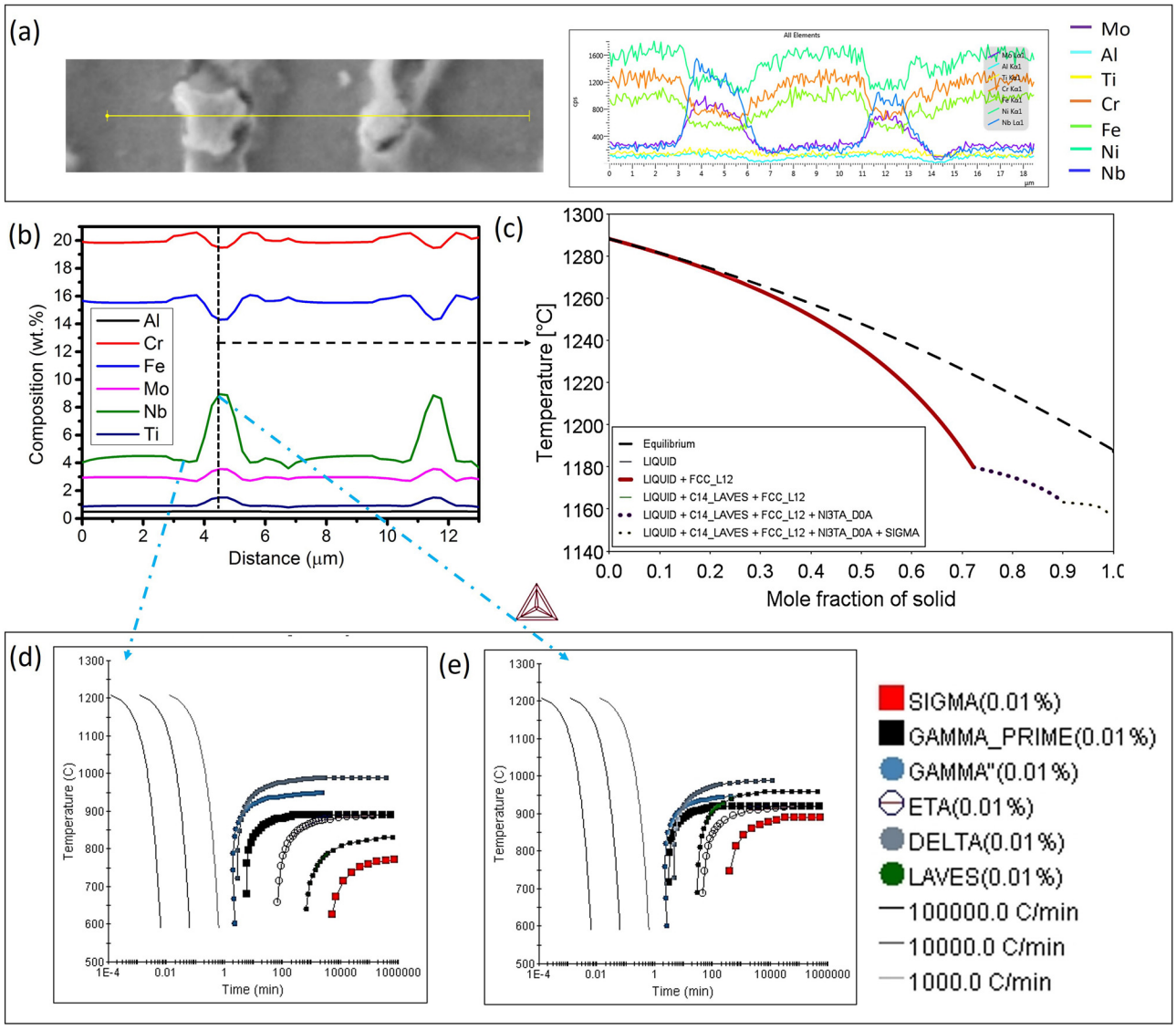


Fig. 6. Effect of segregation (a) experimental EDS line scan in the fusion zone, (b) Virtual EDS line scan in the fusion zone, (c) Scheil's solidification plot for segregated composition, (d) CCT diagram generated by JMat Pro for dendritic core composition, (e) CCT diagram generated by JMat Pro for inter-dendritic composition.

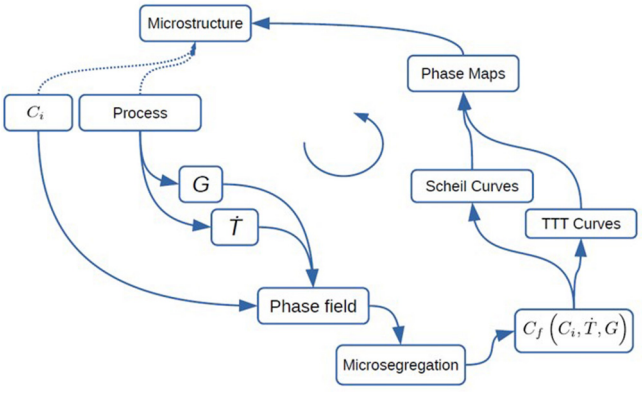


Fig. 7. Integrated framework for the welding process optimization.

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